Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/05/2025
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	11/25/2014	**End Date:**	08/26/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 08/26/2024 per NSSC information (Ed., 11/27/2023) NOTE: End date changed to 08/26/2023 per NSSC information (Ed., 10/13/2022) NOTE: End date changed to 8/26/2022 per NASA GRC (CoInvestigator Stephen Tse has accompanying grant). (Ed., 11/30/2021) NOTE: End date changed to 8/26/2021 due to CoInvestigator Stephen Tse's accompanying grant that ends 8/26/2021 (Ed., 4/20/2020) NOTE: Extended to 11/24/2019 per NSSC info (Ed., 11/25/16)		
Key Personnel Changes/Previous PI:	NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; he will continue reporting.		
COI Name (Institution):	Tse, Stephen Ph.D. (Rutgers University grant 80NSSC18M0134)		
Grant/Contract No.:	NNX15AC79A ; 80NSSC18M0134 (CoI Tse)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Structure and Response of Spherical Diffusion Flames--NNX09AR03A with the same Principal Investigator, Chung K. Law. NOTE April 2020: Stephen Tse, CoInvestigator, continues work under this grant with his grant 80NSSC18M0134, which continues through 8/26/2021; CoI Tse will continue reporting. The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the Glenn Research Center (GRC) and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	The s-Flame experiment aimed to characterize and gain predictive capability of the structure and dynamics of diffusion flames in simple, well-defined flow fields such that the phenomena of interest could be studied without being complicated by various flow field effects. The basic flame configuration adopted was the spherically symmetric diffusion flame generated by discharging a fuel gas from a porous spherical burner into a quiescent, oxidizing ambient in microgravity. The theoretically one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation. The s-Flame experiment used the Combustion Integrated Rack (CIR) and the A dvanced Combustion Microgravity Experiment (ACME) hardware in the Destiny module of the International Space Station (ISS). The s-Flames operations were conducted in two rounds of testing during Increments 60-63, where there was no testing during Increment 62 due to limited crew availability. More specifically, the first round of testing was conducted from Sept. 2019 to Jan. 2020, while the second round of testing – featuring helium rather than nitrogen dilution – occurred from May to July 2020. Nitrogen-diluted methane and ethylene flames in air seemed to demonstrate kinetic extinction at low flow rates (low system Da) and radiative extinction at high flow rates (high system Da). Radiative reabsorption extends flame size at extinction. Surprisingly, radiative extinction was a common event in ISS testing, which was predicted computationally but not previously observed for drop-tower or parabolic flight durations. Similarly, sufficient microgravity duration was needed to observe kinetic extinction because a near-steady flame must first be established before ramping down in flow rate to find the chemical kinetic extinction limit properly. Enriched O2 environments produced smaller characteristic flames; nonetheless, both extinction limits appeared to have been observed. Ethylene flames, which tend to soot under normal gravity conditions, did not soot under microgravity conditions. Soot formation only occurred during forced ignition from the hotwire, but soon dissipated. Inherent sooting was also not found for enriched oxidizer conditions or the inerts, i.e., N2 and He, examined. Nitrogen-diluted hydrogen flames needed very large characteristic sizes to self-extinguish. Radiative extinction was not observed for the cases run, despite lowering the H2 concentration. However, kinetic extinction during ramp-down was exhibited for hydrogen flames. Nitrogen-diluted hydrogen/methane mixture flames manifested radiative extinction, as predicted by computational simulations. However, kinetic extinction at low flow rates could not be identified conclusively. Helium-diluted cases had shifted extinction limits compared to the nitrogen-diluted cases. Ignition was also severely affected by using helium as a diluent. Helium-diluted cases were meant to isolate better pulsating instabilities, theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures above a critical Lewis number. However, at least from the video images, such thermal-diffusive instabilities were not conclusively proven in the ISS experiments, although flame oscillations prior to extinction were manifested. Flame luminosity does not necessarily correspond with flame strength. The highly thermal conductive nature of helium seemed to affect various processes, such as ignition and heat transfer to the burner (affecting isolation of kinetic extinction), despite higher characteristic flame temperatures.
Bibliography Type:	Description: (Last Updated: 11/26/2025)